



## AQ4.2: Activity Questions 2 - Not Graded



This assignment will not be graded and is only for practice.

### Level 1

1 point

Match the vector spaces (with the usual scalar multiplication and vector addition as in  $M_{3 \times 3}(R)$ ) in column A with their bases in column B in Table : M2W4AQ2.

Choose the correct option.

☐

a  $\rightarrow$   $\rightarrow$  ii

☐

a  $\rightarrow$   $\rightarrow$  i

☐

b  $\rightarrow$   $\rightarrow$  iii

☐

c  $\rightarrow$   $\rightarrow$  i

☐

b  $\rightarrow$   $\rightarrow$  ii

☐

c  $\rightarrow$   $\rightarrow$  iii

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

a  $\rightarrow$   $\rightarrow$  ii

b  $\rightarrow$   $\rightarrow$  iii

c  $\rightarrow$   $\rightarrow$  i

If  $S$  be a subset of  $R^5$  such that  $\text{span}(S) = R^5$ , then what is the minimum number of elements possible in  $S$ ?

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 5

1 point

1 point

If  $S_1$  is a maximal linear independent set and  $S_2$  is a minimal spanning set of a vector space  $V$ , then which of the following option(s) is (are) true?

☐

For any  $v \in S_1$ ,  $S_1 \setminus \{v\}$  is a linearly independent.

☐

For any  $v \in S_2$ ,  $S_2 \setminus \{v\}$  is a spanning set of  $V$ .

☐

For any  $v \in V \setminus S_1$ ,  $S_1 \cup \{v\}$  is a linearly dependent.

☐

For any  $v \in V$ ,  $S_2 \cup \{v\}$  is a spanning set of  $V$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**



For any  $v \in S_1$ ,  $v \in S_1$ ,  $S_1 \setminus \{v\}$  is a linearly independent.

For any  $v \in V \setminus S_1$ ,  $v \in V \setminus S_1$ ,  $S_1 \cup \{v\}$  is a linearly dependent.

For any  $v \in V$ ,  $S_2 \cup \{v\}$  is a spanning set of  $V$ .

1 point

Consider the following subset of  $R^2$  with usual addition and scalar multiplication as in  $R^2$ .

- $V_1 = \{(x, 0) \mid x \in R\}$
- $V_2 = \{(0, y) \mid y \in R\}$

Which of the following options are correct?

☐

Both  $V_1$  and  $V_2$  are subspaces of  $R^2$ .

☐

$V_1 \cap V_2$  is a subspace of  $R^2$ .

☐

$V_1 \cup V_2$  is a subspace of  $R^2$ .

☐

$V_1$  is a subspace of  $R^2$ , but  $V_2$  is not.

☐

$V_2$  is a subspace of  $R^2$ , but  $V_1$  is not.

☐

$\{(1, 0)\}$  is a basis of  $V_1$ .

☐

$\{(0, 1)\}$  is a basis of  $V_2$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

Both  $V_1$  and  $V_2$  are subspaces of  $R^2$ .

$V_1 \cap V_2$  is a subspace of  $R^2$ .

$\{(1, 0)\}$  is a basis of  $V_1$ .

$\{(0, 1)\}$  is a basis of  $V_2$ .

1 point

Consider the following subset of  $R^2$  with usual addition and scalar multiplication as in  $R^2$ .

- $V_1 = \{(x, 0) \mid x \in R\}$
- $V_2 = \{(2x, 0) \mid x \in R\}$

Which of the following options are correct?

☐

Both  $V_1$  and  $V_2$  are subspaces of  $R^2$ .

☐

$V_1 \cap V_2$  is a subspace of  $R^2$ .

☐

$V_1 \cup V_2$  is a subspace of  $R^2$ .

☐

$V_1$  is a subspace of  $R^2$ , but  $V_2$  is not.

☐

$V_2$  is a subspace of  $R^2$ , but  $V_1$  is not.

☐

$\{(1, 0)\}$  is a basis of  $V_1$ .

☐

$\{(1, 0)\}$  is a basis of  $V_2$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

Both  $V_1$  and  $V_2$  are subspaces of  $\mathbb{R}^2$ .

$V_1 \cap V_2$  is a subspace of  $\mathbb{R}^2$ .

$V_1 \cup V_2$  is a subspace of  $\mathbb{R}^2$ .

$\{(1, 0)\}$  is a basis of  $V_1$ .

$\{(1, 0)\}$  is a basis of  $V_2$ .

1 point

Let  $V$  be a vector space which is defined as follows:

$$V = \{(x, y, z, w) \mid x + z = y + w\} \subseteq \mathbb{R}^4$$

with usual addition and scalar multiplication. Which of the following set forms a basis of  $V$ ?

☐

$\{(1, 0, 0, 0), (0, 1, 0, 0), (0, 0, 0, 1)\}$

☐

$\{(1, 1, 0, 0), (0, 1, -1, 0), (0, -1, 0, 1)\}$

☐

$\{(1, 0, -1, 0), (1, 1, 0, 0), (1, 0, 0, 1)\}$

☐

$\{(1, 1, 0, 0), (0, 1, 1, 0), (0, -1, 0, 1)\}$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$\{(1, 0, -1, 0), (1, 1, 0, 0), (1, 0, 0, 1)\}$

$\{(1, 1, 0, 0), (0, 1, 1, 0), (0, -1, 0, 1)\}$

**Level 2**

1 point

Which of the following sets form a basis of the vector space of  $2 \times 2$  lower triangular real matrices with usual matrix addition and scalar multiplication? (More than one option may be correct)

☐

$\left\{\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}\right\}$

☐

$\left\{\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right\}$

☐

$\left\{\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}\right\}$

☐

$\left\{\begin{bmatrix} -1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}\right\}$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$\left\{\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}\right\}$

$\left\{\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right\}$

$\left\{\begin{bmatrix} -1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}\right\}$

If  $SS$  is a linearly independent set in vector space  $V = \{(x, y) \mid y = 2x, \text{ where } x, y \in \mathbb{R}\}$  with usual addition and scalar multiplication as in  $\mathbb{R}^2$ , then find the maximum possible cardinality of  $SS$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 1

1 point

1 point

If  $\{v_1, v_2, v_3\}$  forms a basis of  $\mathbb{R}^3$ , then which of the following are true?

☐

$\{v_1, v_2, v_1 + v_3\}$  forms a basis of  $\mathbb{R}^3$

☐

$\{v_1, v_1 + v_2, v_1 + v_3\}$  forms a basis of  $\mathbb{R}^3$ .

☐

$\{v_1, v_1 + v_2, v_1 - v_3\}$  forms a basis of  $\mathbb{R}^3$ .

☐

$\{v_1, v_1 - v_2, v_1 - v_3\}$  forms a basis of  $\mathbb{R}^3$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$\{v_1, v_2, v_1 + v_3\}$  forms a basis of  $\mathbb{R}^3$

$\{v_1, v_1 + v_2, v_1 + v_3\}$  forms a basis of  $\mathbb{R}^3$ .

$\{v_1, v_1 + v_2, v_1 - v_3\}$  forms a basis of  $\mathbb{R}^3$ .

$\{v_1, v_1 - v_2, v_1 - v_3\}$  forms a basis of  $\mathbb{R}^3$ .

1 point

Which of the following options is(are) true?

☐

Any minimal spanning set of a vector space  $V$  must be a basis of  $V$ .

☐

Any maximal spanning set of a vector space  $V$  must be a basis of  $V$ .

☐

Any minimal linear independent set of vector space  $V$  must be a basis of  $V$ .

☐

Any maximal linear independent set of vector space  $V$  must be a basis of  $V$ .

☐ The basis of a vector space is unique.

☐

The number of elements in a basis of  $\mathbb{R}^3$  is 33.

☐

The number of elements in a basis of  $M_{3 \times 3}(\mathbb{R})$  is 33.

☐

There are infinite number of bases of  $\mathbb{R}^3$ .

☐

Any subset of a minimal spanning set of  $V$  cannot be a spanning set.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

Any minimal spanning set of a vector space  $V$  must be a basis of  $V$ .

Any maximal linear independent set of vector space  $V$  must be a basis of  $V$ .

The number of elements in a basis of  $\mathbb{R}^3$  is 33.

There are infinite number of bases of  $R^3$ .

Any subset of a minimal spanning set of  $V$  cannot be a spanning set.

Check Answers

Your score is: 0/10